Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ''. Analysis was performed against Reactome version 93 on 06/09/2025. The web link to these results is:

https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTA5MDYxMjUxMDJfMzM4Njg%3D

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

Table of Contents

1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome’s team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini–Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

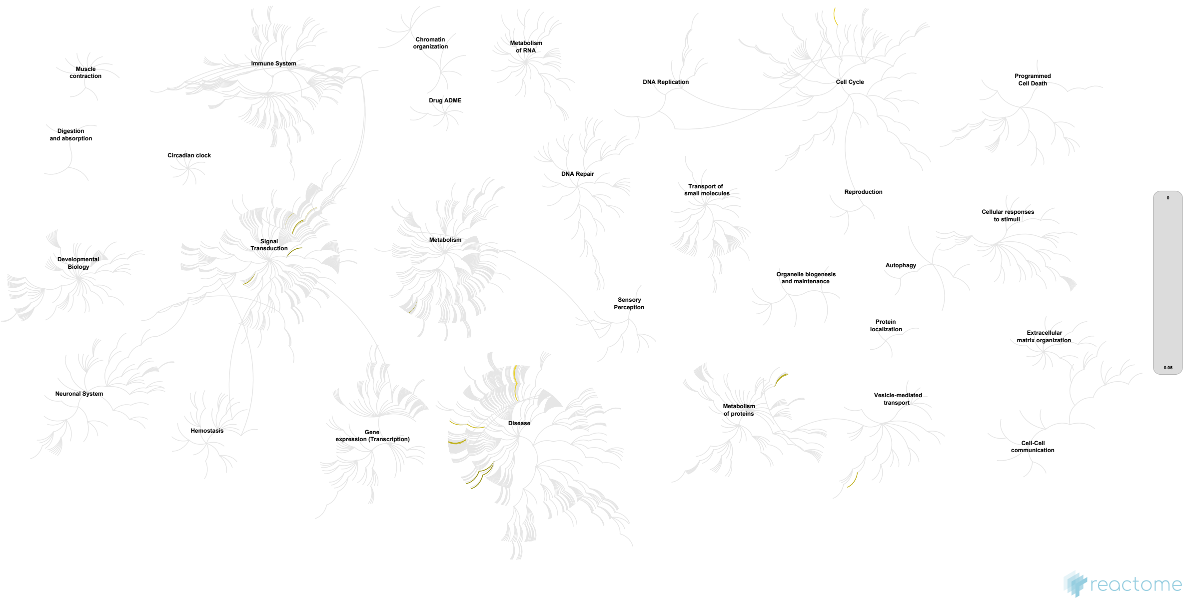
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, … D’Eustachio P (2016). The reactome pathway knowledgebase. Nucleic Acids Research, 44(D1), D481–D487. https://doi.org/10.1093/nar/gkv1351.

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, … Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. BMC Bioinformatics, 18.

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamani-Hochberg method.
- 10 out of 18 identifiers in the sample were found in Reactome, where 612 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent.
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTA5MDYxMjUxMDJfMzM4Njg%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Loss of Function of SMAD4 in Cancer	1 / 3	1.22e-04	0.004	0.344	1 / 1	6.43e-05
SMAD4 MH2 Domain Mutants in Cancer	1 / 3	1.22e-04	0.004	0.344	1 / 1	6.43e-05
Drug-mediated inhibition of CDK4/CDK6 activity	1 / 6	2.43e-04	0.008	0.344	2 / 2	1.29e-04
TGFBR1 KD Mutants in Cancer	1 / 6	2.43e-04	0.008	0.344	1 / 1	6.43e-05
SMAD2/3 Phosphorylation Motif Mutants in Cancer	1 / 7	2.84e-04	0.009	0.344	1 / 1	6.43e-05
Loss of Function of TGFBR1 in Cancer	1 / 7	2.84e-04	0.009	0.344	1 / 2	1.29e-04
Activated NTRK2 signals through PI3K	1 / 11	4.46e-04	0.014	0.344	2 / 2	1.29e-04
Defective EXT2 causes exostoses 2	1 / 16	6.48e-04	0.021	0.344	3 / 3	1.93e-04
Defective EXT1 causes exostoses 1, TRPS2 and CHDS	1 / 16	6.48e-04	0.021	0.344	3 / 3	1.93e-04
MET activates RAP1 and RAC1	1 / 17	6.89e-04	0.022	0.344	5 / 5	3.21e-04
Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)	1 / 17	6.89e-04	0.022	0.344	1 / 1	6.43e-05
Aberrant regulation of mitotic G1/S transition in cancer due to RB1 defects	1 / 17	6.89e-04	0.022	0.344	1 / 2	1.29e-04
Defective POMGNT1 causes MDDGA3, MDDGB3 and MDDGC3	1 / 18	7.30e-04	0.023	0.344	1 / 1	6.43e-05
Defective POMT1 causes MDDGA1, MDDGB1 and MDDGC1	1 / 19	7.70e-04	0.024	0.344	1 / 1	6.43e-05
Defective POMT2 causes MDDGA2, MDDGB2 and MDDGC2	1 / 19	7.70e-04	0.024	0.344	1 / 1	6.43e-05
Defective B3GALT6 causes EDSP2 and SEMDJL1	1 / 21	8.51e-04	0.027	0.344	1 / 1	6.43e-05
Defective B4GALT7 causes EDS, progeroid type	1 / 21	8.51e-04	0.027	0.344	1 / 1	6.43e-05
Formation of annular gap junctions	1 / 21	8.51e-04	0.027	0.344	1 / 2	1.29e-04
MET receptor recycling	1 / 22	8.92e-04	0.028	0.344	2 / 2	1.29e-04
Defective B3GAT3 causes JDSSDHD	1 / 22	8.92e-04	0.028	0.344	1 / 1	6.43e-05
PI3K events in ERBB2 signaling	1 / 22	8.92e-04	0.028	0.344	3 / 7	4.50e-04
DAG1 core M1 glycosylations	1 / 25	0.001	0.032	0.344	2 / 2	1.29e-04
MET activates PI3K/AKT signaling	1 / 26	0.001	0.033	0.344	5 / 5	3.21e-04

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
DAG1 core M2 glycosylations	1 / 26	0.001	0.033	0.344	3 / 3	1.93e-04
Erythropoietin activates Phosphoinositide-3-kinase (PI3K)	1 / 29	0.001	0.037	0.344	4 / 7	4.50e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Loss of Function of SMAD4 in Cancer (R-HSA-3304347)

Diseases: cancer.

SMAD4 was identified as a gene homozygously deleted in ~30% of pancreatic cancers and was named DPC4 (DPC stands for deleted in pancreatic cancer). SMAD4 maps to the chromosomal band 18q21.1, and about 90% of pancreatic carcinomas show allelic loss at chromosomal arm 18q (Hahn et al. 1996), while ~50% of pancreatic cancers show some alteration of the SMAD4 gene (reviewed by Schutte et al. 1999).

Based on COSMIC database (Catalogue Of Somatic Mutations In Cancer) (Forbes et al. 2011), mutations in the coding sequence of SMAD4 gene are frequently found in pancreatic cancer, biliary duct carcinoma and colorectal cancer (reviewed by Schutte et al. 1999). Germline SMAD4 mutations are the cause of juvenile polyposis, an autosomal dominant disease that predisposes affected individuals to hamartomatous polyps and gastrointestinal cancer (Howe et al. 1998). Homozygous Smad4 loss is embryonic lethal in mice (Takaku et al. 1998). Smad4 +/- heterozygotes appear normal but develop intestinal polyps between 6 and12 months of age and these polyps can progress to cancer. Loss of the remaining wild-type Smad4 allele is detectable only at later stages of tumor progression in Smad4+/- mice (Xu et al. 2000). Compound Apc+/-;Smad4+/- mice develop malignant tumors from intestinal polyps more rapidly than Apc+/- mice (Takaku et al. 1998).

SMAD4 coding sequence mutations are most frequently found in the MH2 domain and impair the formation of SMAD4 heterotrimers with phosphorylated SMAD2 and SMAD3 (Shi et al. 1997, Fleming et al. 2013), thereby impairing SMAD4:SMAD2/3 heterotrimer-mediated transcriptional regulation of TGF-beta responsive genes. MH2 domain is also involved in the formation of SMAD4 homotrimers which may play a role in SMAD4 protein stability (Shi et al. 1997).

Coding sequence mutations are also found in the MH1 domain of SMAD4. MH1 domain is involved in DNA binding (Dai et al. 1999) and it is also involved in the formation of SMAD4 homotrimers (Hata et al. 1997).

References

Deng CX, Kim SJ, Xu X, Im YH, Brodie SG, Chen L, ... Zhou YX (2000). Haploid loss of the tumor suppressor Smad4/Dpc4 initiates gastric polyposis and cancer in mice. Oncogene, 19, 1868-74.

Matsui M, Takaku K, Miyoshi H, Oshima M, Seldin MF & Taketo MM (1998). Intestinal tumorigenesis in compound mutant mice of both Dpc4 (Smad4) and Apc genes. Cell, 92, 645-56.

Lo RS, Hata A, Massagué J, Lagna G & Wotton D (1997). Mutations increasing autoinhibition inactivate tumour suppressors Smad2 and Smad4. Nature, 388, 82-7.

Aaltonen LA, Houlston RS, Summers RW, Ringold JC, Stone EM, Järvinen HJ, ... Roth S (1998). Mutations in the SMAD4/DPC4 gene in juvenile polyposis. Science, 280, 1086-8.

Pavletich NP, Lo RS, Hata A, Shi Y & Massagué J (1997). A structural basis for mutational inactivation of the tumour suppressor Smad4. Nature, 388, 87-93.

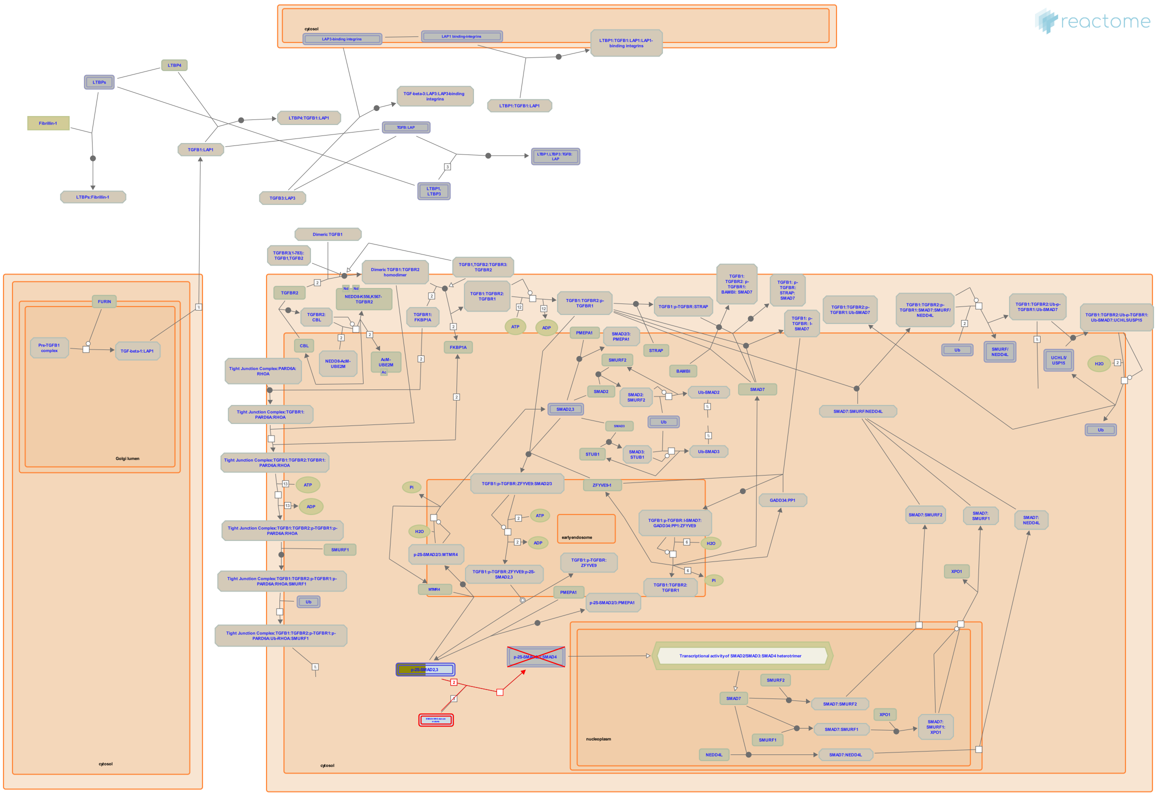
Edit history

Date	Action	Author
2013-04-23	Created	Orlic-Milacic M
2013-05-03	Edited	Jassal B
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M
2023-10-12	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SMAD2	Q15796

2. SMAD4 MH2 Domain Mutants in Cancer (R-HSA-3311021)



Diseases: cancer.

The MH2 domain of SMAD4 is the most frequently mutated SMAD4 region in cancer. MH2 domain mutations result in the loss of function of SMAD4 by abrogating the formation of transcriptionally active heterotrimers of SMAD4 and TGF-beta receptor complex-activated R-SMADs - SMAD2 and SMAD3 (Shi et al. 1997, Chacko et al. 2001, Chacko et al. 2004, Fleming et al. 2013).

The hotspot MH2 domain amino acid residues that are targeted by missense mutations are Asp351 (D351), Pro356 (P356) and Arg361 (R361). These three hotspot residues map to the L1 loop which is conserved in SMAD2 and SMAD3 and is involved in intermolecular interactions that contribute to the formation of SMAD heterotrimers and homotrimers (Shi et al. 1997, Fleming et al. 2013). Other frequently mutated residues in the MH2 domain of SMAD4 - Ala406 (A406), Lys428 (K428) and Arg515 (R515) - are involved in binding the phosphorylation motif (Ser-Ser-X-Ser) of SMAD2 and SMAD3, with Arg515 in the L3 loop being critical for this interaction (Chacko et al. 2001, Chacko et al. 2004, Fleming et al. 2013).

References

Shi G, De Caestecker M, Lin K, Chacko BM, Hayward LJ, Tiwari A, ... Lam S (2004). Structural basis of heteromeric smad protein assembly in TGF-beta signaling. Mol Cell, 15, 813-23.

Pavletich NP, Lo RS, Hata A, Shi Y & Massagué J (1997). A structural basis for mutational inactivation of the tumour suppressor Smad4. Nature, 388, 87-93.

Correia JJ, Lam SS, de Caestecker MP, Qin B, Chacko BM & Lin K (2001). The L3 loop and C-terminal phosphorylation jointly define Smad protein trimerization. Nat. Struct. Biol., 8, 248-53.

Mouradov D, Jorissen RN, Jones IT, Tsui C, Palmieri M, Sieber OM, ... Zhao Q (2013). SMAD2, SMAD3 and SMAD4 mutations in colorectal cancer. Cancer Res., 73, 725-35.

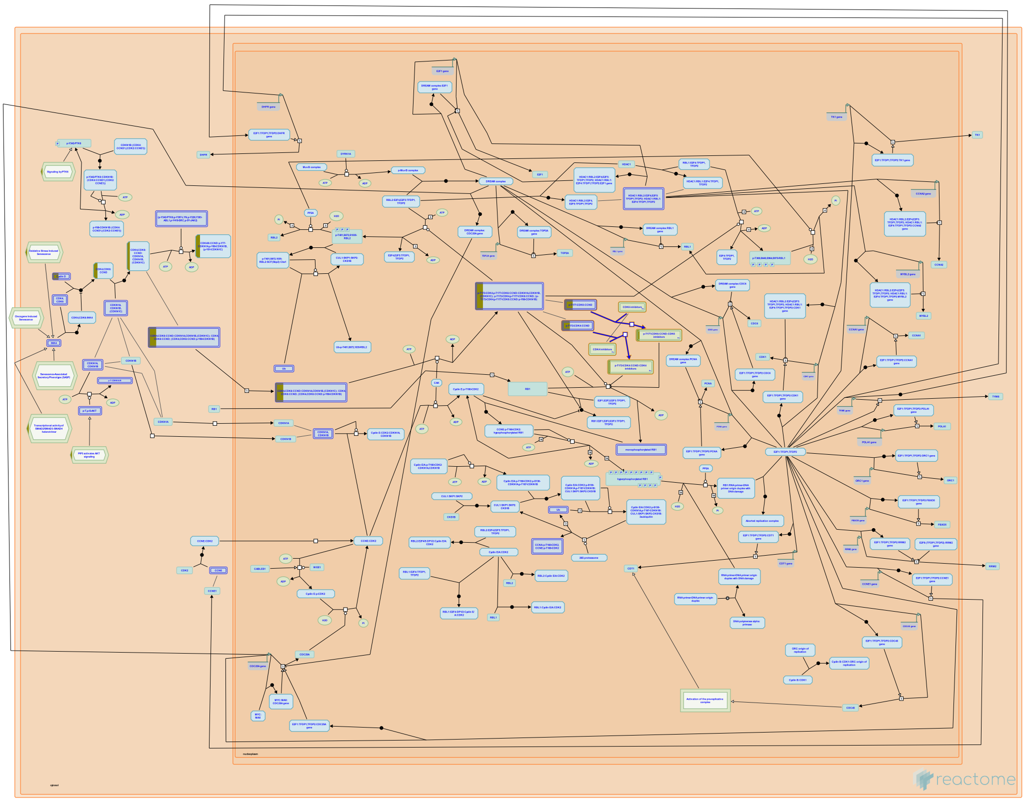
Edit history

Date	Action	Author
2013-04-24	Created	Orlic-Milacic M
2013-05-03	Edited	Jassal B
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SMAD2	Q15796

3. Drug-mediated inhibition of CDK4/CDK6 activity (R-HSA-9754119)



Cyclin dependent kinases CDK4 and CDK6 regulate crucial steps in the G1 phase of the cell cycle that commit cells to transition to the S phase and ultimately divide. Many growth signaling pathways, frequently perturbed in cancer, converge on CDK4/CDK6 activation, thus driving cellular proliferation. This makes CDK4 and CDK6 promising targets for anti-cancer therapy. So far, three CDK4/6 inhibitors, palbociclib, ribociclib and abemaciclib, have been approved for clinical use and many others are at different stages of clinical testing. CDK4/6 inhibitors mainly have a cytostatic effect on tumor cells, but can also influence immune response to tumor by targeting immune system cells in the tumor microenvironment. While intact RB1, the main target of CDK4/6 during cell cycle progression, is in general considered to be a prerequisite for the success of CDK4/6-targeted anti-cancer therapy, the status of other, less explored CDK4/6 targets can also affect the treatment outcome. For review, please refer to Asghar et al. 2015, Klein et al. 2018, Álvarez-Fernández and Malumbres 2020, Petroni et al. 2020).

References

Witkiewicz AK, Asghar U, Turner NC & Knudsen ES (2015). The history and future of targeting cyclin-dependent kinases in cancer therapy. Nat Rev Drug Discov, 14, 130-46.

Petroni G, Formenti SC, Chen-Kiang S & Galluzzi L (2020). Immunomodulation by anticancer cell cycle inhibitors. Nat Rev Immunol, 20, 669-679.

Tap WD, Davis LE, Koff A, Kovatcheva M & Klein ME (2018). CDK4/6 Inhibitors: The Mechanism of Action May Not Be as Simple as Once Thought. Cancer Cell, 34, 9-20.

Alvarez-Fernández M & Malumbres M (2020). Mechanisms of Sensitivity and Resistance to CDK4/6 Inhibition. Cancer Cell, 37, 514-529.

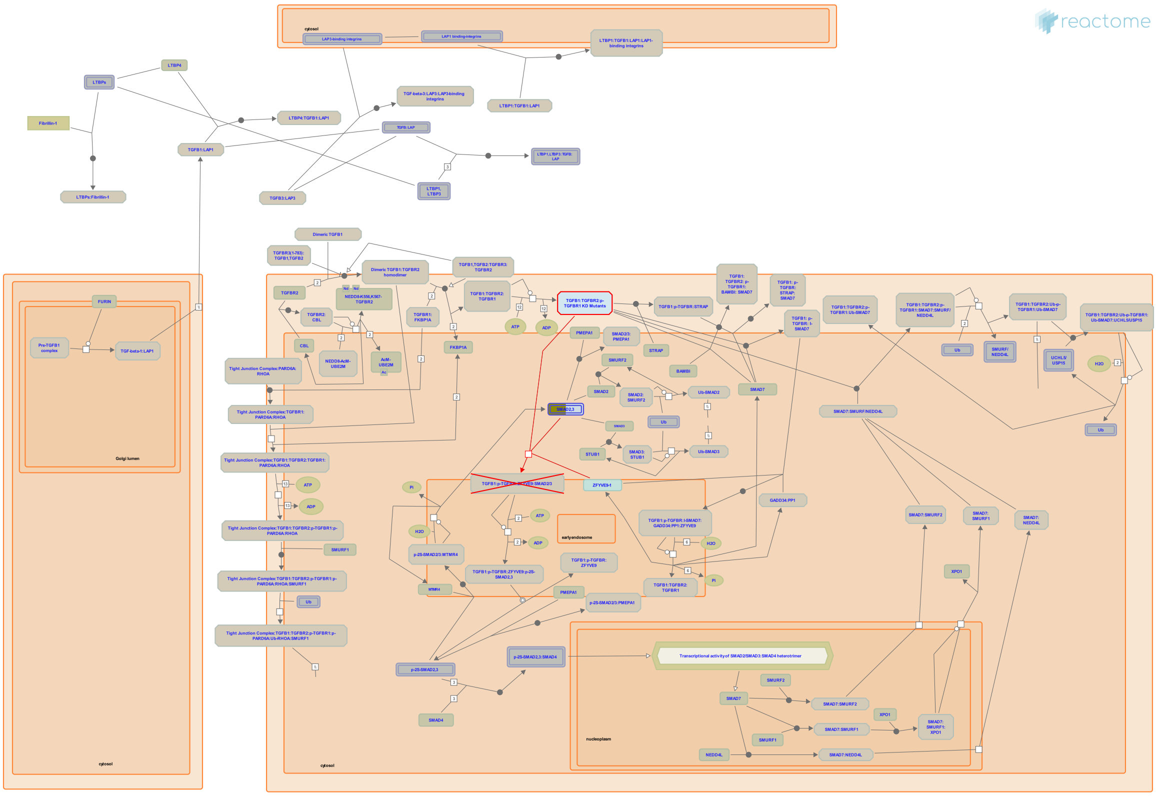
Edit history

Date	Action	Author
2021-09-21	Authored	Orlic-Milacic M
2021-09-21	Created	Orlic-Milacic M
2021-11-02	Reviewed	Kadambat Nair S
2021-11-12	Edited	Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
CCND2	P30279

4. TGFBR1 KD Mutants in Cancer (R-HSA-3656532)



Diseases: cancer.

Mutations in the kinase domain (KD) of TGF-beta receptor 1 (TGFBR1) have been found in Ferguson-Smith tumor i.e. multiple self-healing squamous epithelioma - MSSE (Goudie et al. 2011), breast cancer (Chen et al. 1998), ovarian cancer (Chen et al. 2001) and head-and-neck cancer (Chen et al. 2001). KD mutations reported in MSSE are nonsense and frameshift mutations that cause premature termination of TGFBR1 translation, resulting in truncated receptors that lack substantial portions of the kinase domain, or cause nonsense-mediated decay of mutant transcripts. A splice site KD mutation c.806-2A>C is predicted to result in the skipping of exon 5 and the absence of KD amino acid residues 269-324 from the mutant receptor. The splice site mutant is expressed at the cell surface but unresponsive to TGF-beta stimulation (Goudie et al. 2004).

TGFBR1 KD mutations reported in breast, ovarian and head-and-neck cancer are missense mutations, and it appears that these mutant proteins are partially functional but that their catalytic activity or protein stability is decreased (Chen et al. 1998, Chen et al. 2001a and b). These mutants are not shown.

References

McNiff J, Leffell D, Chen T, Rimm DL, Wells RG, Yan W & Reiss M (2001). Novel inactivating mutations of transforming growth factor-beta type I receptor gene in head-and-neck cancer metastases. Int. J. Cancer, 93, 653-61.

Gerdes AM, Reversade B, Lee H, Ferguson-Smith MA, Whittaker S, Christie L, ... Verma C (2011). Multiple self-healing squamous epithelioma is caused by a disease-specific spectrum of mutations in TGFBR1. Nat. Genet., 43, 365-9.

Garrigue-Antar L, Chen T, Reiss M & Carter D (1998). Transforming growth factor beta type I receptor kinase mutant associated with metastatic breast cancer. Cancer Res., 58, 4805-10.

Colligan B, Chen T, Graff JR, Hurst B, Dehner B, Pemberton J, ... Triplett J (2001). Transforming growth factor-beta receptor type I gene is frequently mutated in ovarian carcinomas. Cancer Res., 61, 4679-82.

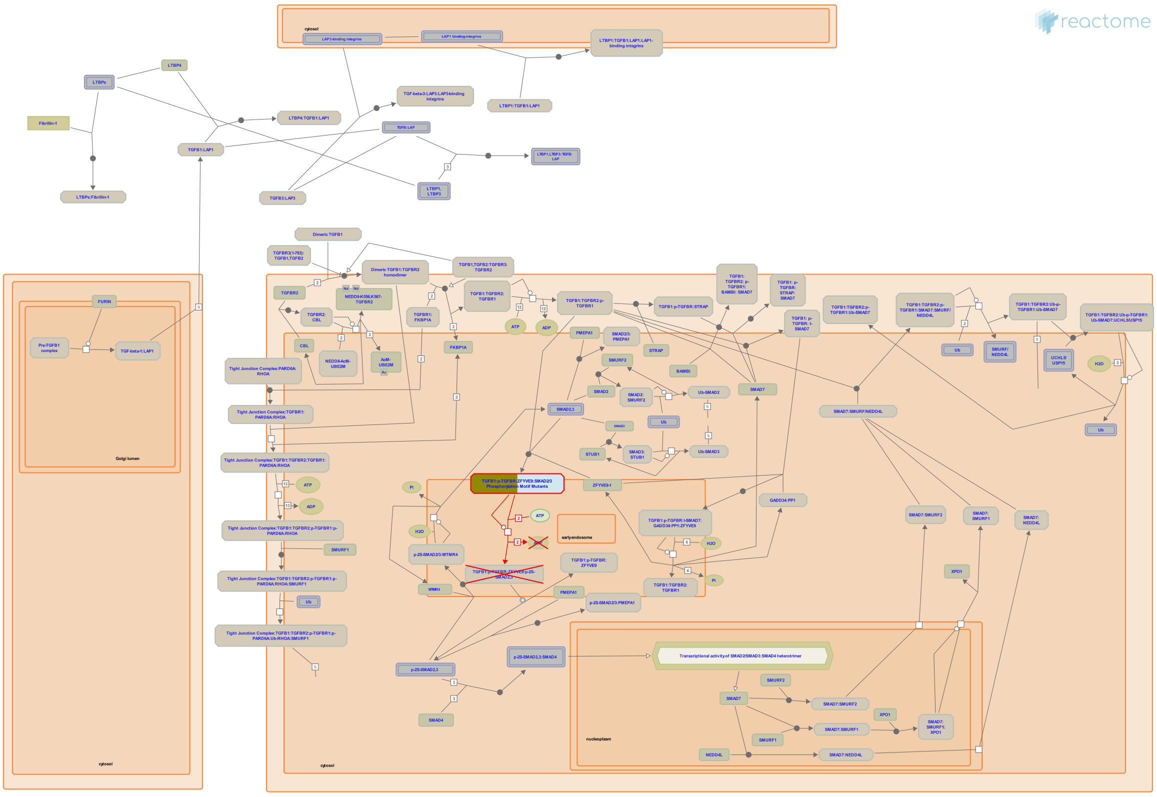
Edit history

Date	Action	Author
2013-06-05	Created	Orlic-Milacic M
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SMAD2	Q15796

5. SMAD2/3 Phosphorylation Motif Mutants in Cancer (R-HSA-3304356)



Diseases: cancer.

The conserved phosphorylation motif Ser-Ser-X-Ser at the C-terminus of SMAD2 and SMAD3 is subject to disruptive mutations in cancer. The last two serine residues in this conserved motif, namely Ser465 and Ser467 in SMAD2 and Ser423 and Ser425 in SMAD3, are phosphorylated by the activated TGF beta receptor complex (Macias Silva et al. 1996, Nakao et al. 1997). Once phosphorylated, SMAD2 and SMAD3 form transcriptionally active heterotrimers with SMAD4 (Chacko et al. 2001, Chacko et al. 2004). Phosphorylation motif mutants of SMAD2 and SMAD3 cannot be activated by the TGF-beta receptor complex either because serine residues are substituted with amino acid residues that cannot be phosphorylated or because the phosphorylation motif is deleted from the protein sequence or truncated (Fleming et al. 2013).

References

Shi G, De Caestecker M, Lin K, Chacko BM, Hayward LJ, Tiwari A, ... Lam S (2004). Structural basis of heteromeric smad protein assembly in TGF-beta signaling. Mol Cell, 15, 813-23.

Wrana JL, Attisano L, Abdollah S, Hoodless PA, Pirone R & Macias-Silva M (1996). MADR2 is a substrate of the TGFbeta receptor and its phosphorylation is required for nuclear accumulation and signaling. Cell, 87, 1215-24.

Correia JJ, Lam SS, de Caestecker MP, Qin B, Chacko BM & Lin K (2001). The L3 loop and C-terminal phosphorylation jointly define Smad protein trimerization. Nat. Struct. Biol., 8, 248-53.

Souchelnytskyi S, Engstrom U, ten Dijke P, Heldin CH, Wernstedt C & Tamaki K (1997). Phosphorylation of Ser465 and Ser467 in the C terminus of Smad2 mediates interaction with Smad4 and is required for transforming growth factor-beta signaling. J Biol Chem, 272, 28107-15.

Mouradov D, Jorissen RN, Jones IT, Tsui C, Palmieri M, Sieber OM, ... Zhao Q (2013). SMAD2, SMAD3 and SMAD4 mutations in colorectal cancer. Cancer Res., 73, 725-35.

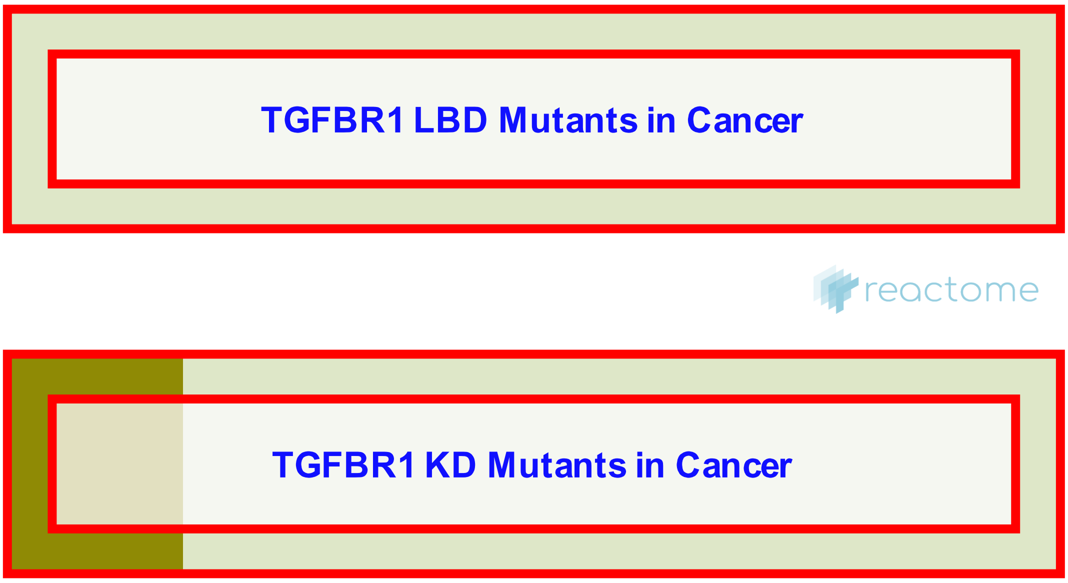
Edit history

Date	Action	Author
2013-04-23	Created	Orlic-Milacic M
2013-05-03	Edited	Jassal B
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SMAD2	Q15796

6. Loss of Function of TGFBR1 in Cancer (R-HSA-3656534)



Diseases: cancer.

TGF-beta receptor 1 (TGFBR1) loss-of-function is a less frequent mechanism for inactivation of TGF-beta signaling in cancer compared to SMAD4 and TGFBR2 inactivation. Genomic deletion of TGFBR1 locus has been reported in pancreatic cancer (Goggins et al. 1998), biliary duct cancer (Goggins et al. 1998) and lymphoma (Schiemann et al. 1999), while loss-of-function mutations have been reported in breast (Chen et al. 1998) and ovarian cancer (Chen et al. 2001), metastatic head-and-neck cancer (Chen et al. 2001), and in Ferguson-Smith tumors (multiple self-healing squamous epithelioma - MSSE) (Goudie et al. 2011). Loss-of-function mutations mainly affect the ligand-binding extracellular domain of TGFBR1 and the kinase domain of TGFBR1 (Goudie et al. 2011). In the mouse model of colorectal cancer, Tgfbr1 haploinsufficiency cooperates with Apc haploinsufficiency in the development of intestinal tumors (Zeng et al. 2009).

References

McNiff J, Leffell D, Chen T, Rimm DL, Wells RG, Yan W & Reiss M (2001). Novel inactivating mutations of transforming growth factor-beta type I receptor gene in head-and-neck cancer metastases. Int. J. Cancer, 93, 653-61.

Gerdes AM, Reversade B, Lee H, Ferguson-Smith MA, Whittaker S, Christie L, ... Verma C (2011). Multiple self-healing squamous epithelioma is caused by a disease-specific spectrum of mutations in TGFBR1. Nat. Genet., 43, 365-9.

Garrigue-Antar L, Chen T, Reiss M & Carter D (1998). Transforming growth factor beta type I receptor kinase mutant associated with metastatic breast cancer. Cancer Res., 58, 4805-10.

Phukan S, Zeng Q, Pasche B, Yang GY, Liao J, Xu Y, ... Sadim M (2009). Tgfbr1 haploinsufficiency is a potent modifier of colorectal cancer development. Cancer Res., 69, 678-86.

Colligan B, Chen T, Graff JR, Hurst B, Dehner B, Pemberton J, ... Triplett J (2001). Transforming growth factor-beta receptor type I gene is frequently mutated in ovarian carcinomas. Cancer Res., 61, 4679-82.

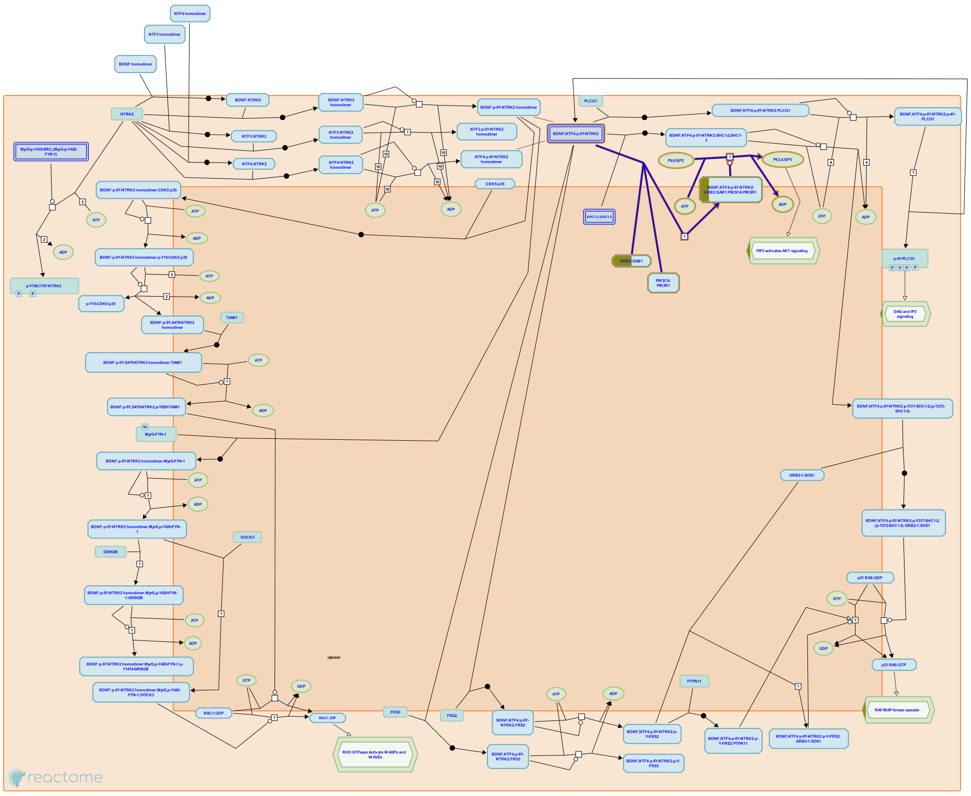
Edit history

Date	Action	Author
2013-06-05	Created	Orlic-Milacic M
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M
2023-10-12	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SMAD2	Q15796

7. Activated NTRK2 signals through PI3K (R-HSA-9028335)



Neurotrophin receptor NTRK2 (TRKB), activated by BDNF or NTF4, activates PI3K, resulting in formation of the PIP3 secondary messenger. PIP3 activates AKT signaling, and AKT signaling activates mTOR signaling (Yuen and Mobley 1999, Cao et al. 2013).

References

Marshall J, Spaller MR, Rioult-Pedotti MS, Migani P, Parang K, Cao C, ... Tiwari R (2013). Impairment of TrkB-PSD-95 signaling in Angelman syndrome. PLoS Biol., 11, e1001478.

Yuen EC & Mobley WC (1999). Early BDNF, NT-3, and NT-4 signaling events. Exp. Neurol., 159, 297-308.

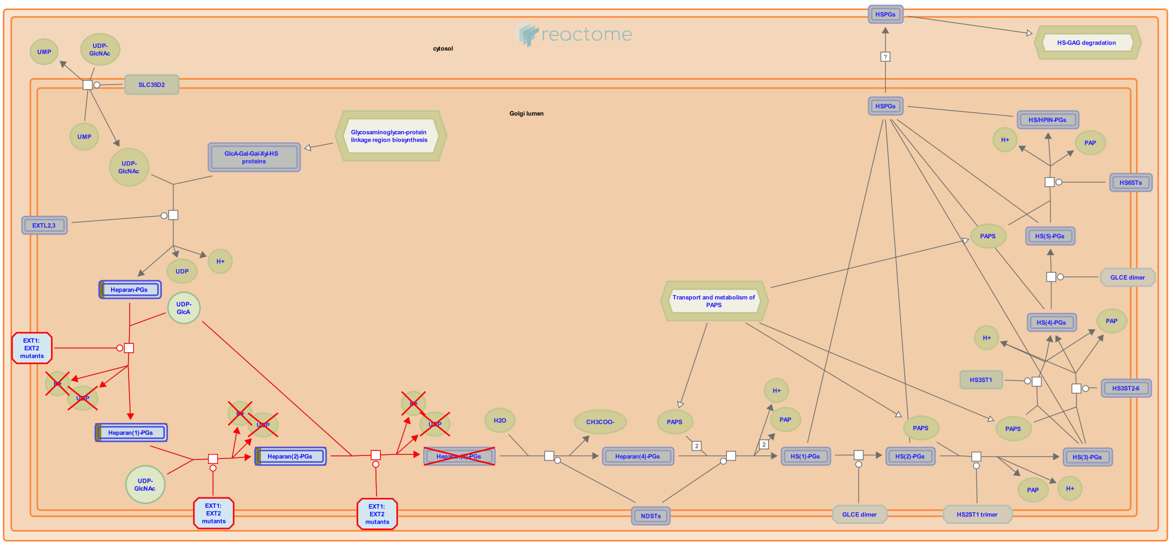
Edit history

Date	Action	Author
2017-11-07	Created	Orlic-Milacic M
2018-01-05	Authored	Orlic-Milacic M
2018-02-13	Reviewed	Antila H, Castrén E
2018-02-20	Edited	Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GAB1	Q13480

8. Defective EXT2 causes exostoses 2 (R-HSA-3656237)



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterised by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands.

References

Peterson HA (1989). Multiple hereditary osteochondromata. Clin. Orthop. Relat. Res., 222-30.

Evans DG, McGaughran JM & Ward HB (1995). WAGR syndrome and multiple exostoses in a patient with del(11)(p11.2p14.2). J. Med. Genet., 32, 823-4.

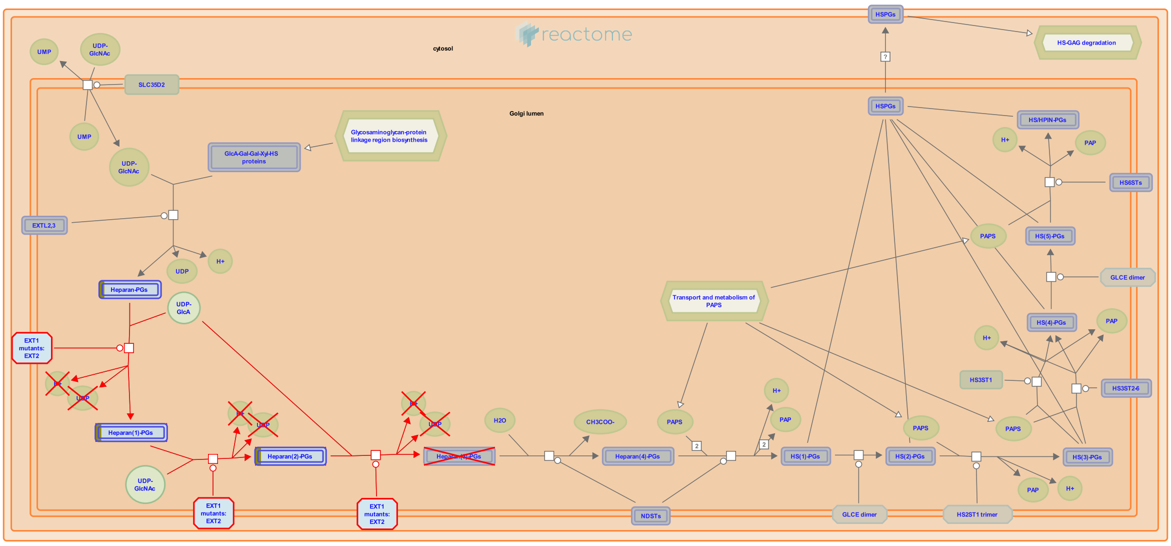
Edit history

Date	Action	Author
2013-05-31	Edited	Jassal B
2013-05-31	Authored	Jassal B
2013-05-31	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GPC6	Q9Y625

9. Defective EXT1 causes exostoses 1, TRPS2 and CHDS (R-HSA-3656253)



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterized by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands. Trichorhinophalangeal syndrome, type II (TRPS2 aka Langer-Giedion syndrome, LGS) is a disorder that combines the clinical features of trichorhinophalangeal syndrome type I (TRPS1, MIM:190350) and multiple exostoses type I, caused by mutations in the TRPS1 and EXT1 genes, respectively (Langer et al. 1984, Ludecke et al. 1995). Defects in EXT1 may also be responsible for chondrosarcoma (CHDS; MIM:215300) (Schajowicz & Bessone 1967, Hecht et al. 1995).

References

Parrish JE, Willems PJ, Wagner MJ, Nardmann J, Frydman M, Wells DE, ... La Pillo B (1995). Molecular dissection of a contiguous gene syndrome: localization of the genes involved in the Langer-Giedion syndrome. Hum. Mol. Genet., 4, 31-6.

Bessone JE & Schajowicz F (1967). Chondrosarcoma in three brothers. A pathological and genetic study. J Bone Joint Surg Am, 49, 129-41.

Peterson HA (1989). Multiple hereditary osteochondromata. Clin. Orthop. Relat. Res., 222-30.

Lutter LD, Jennings CG, Krassikoff N, Day DW, Langer LO, Scheer-Williams M, ... Gorlin RJ (1984). The tricho-rhino-phalangeal syndrome with exostoses (or Langer-Giedion syndrome): four additional patients without mental retardation and review of the literature. Am. J. Med. Genet., 19, 81-112.

Evans DG, McGaughran JM & Ward HB (1995). WAGR syndrome and multiple exostoses in a patient with del(11)(p11.2p14.2). J. Med. Genet., 32, 823-4.

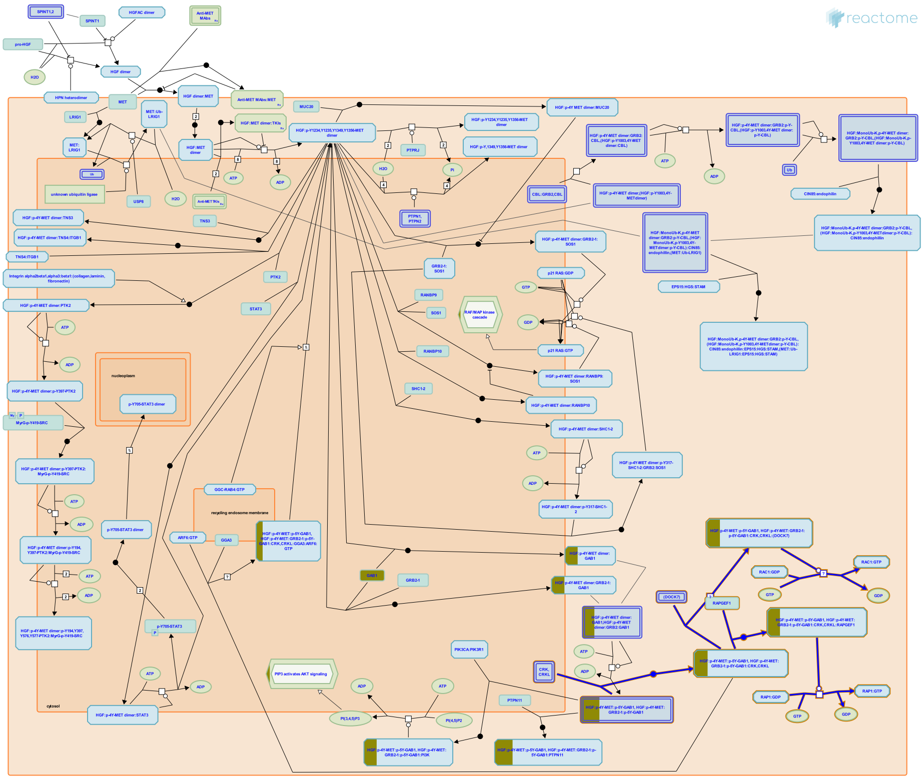
Edit history

Date	Action	Author
2013-05-31	Edited	Jassal B
2013-05-31	Authored	Jassal B
2013-05-31	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GPC6	Q9Y625

10. MET activates RAP1 and RAC1 (R-HSA-8875555)



The adapter protein GAB1 is involved in recruitment, through CRK and related CRKL proteins, of guanyl nucleotide exchange factors (GEFs) to the activated MET receptor. MET-associated GEFs, such as RAPGEF1 (C3G) and DOCK7, activate RAP1 and RAC1, respectively, leading to morphological changes that contribute to cell motility (Schaeper et al. 2000, Sakkab et al. 2000, Lamorte et al. 2002, Watanabe et al. 2006, Murray et al. 2014).

References

Kempkes B, Fuchs KP, Birchmeier W, Sachs M, Gehring NH & Schaeper U (2000). Coupling of Gab1 to c-Met, Grb2, and Shp2 mediates biological responses. J. Cell Biol., 149, 1419-32.

van Aelst L, Didier S, Byrne AT, Murray DW, Paulino V, Ruggieri R, ... Symons M (2014). Guanine nucleotide exchange factor Dock7 mediates HGF-induced glioblastoma cell invasion via Rac activation. Br. J. Cancer, 110, 1307-15.

Lewitzky M, Sakkab D, Posern G, Feller SM, Birchmeier W, Sachs M & Schaeper U (2000). Signaling of hepatocyte growth factor/scatter factor (HGF) to the small GTPase Rap1 via the large docking protein Gab1 and the adapter protein CRKL. J. Biol. Chem., 275, 10772-8.

Tanaka S, Makino Y, Ichihara S, Tsuda M, Watanabe T, Mochizuki N, ... Nagashima K (2006). Adaptor molecule Crk is required for sustained phosphorylation of Grb2-associated binder 1 and hepatocyte growth factor-induced cell motility of human synovial sarcoma cell lines. Mol. Cancer Res., 4, 499-510.

Lamorte L, Naujokas M, Park M & Royal I (2002). Crk adapter proteins promote an epithelial-mesenchymal-like transition and are required for HGF-mediated cell spreading and breakdown of epithelial adherens junctions. Mol. Biol. Cell, 13, 1449-61.

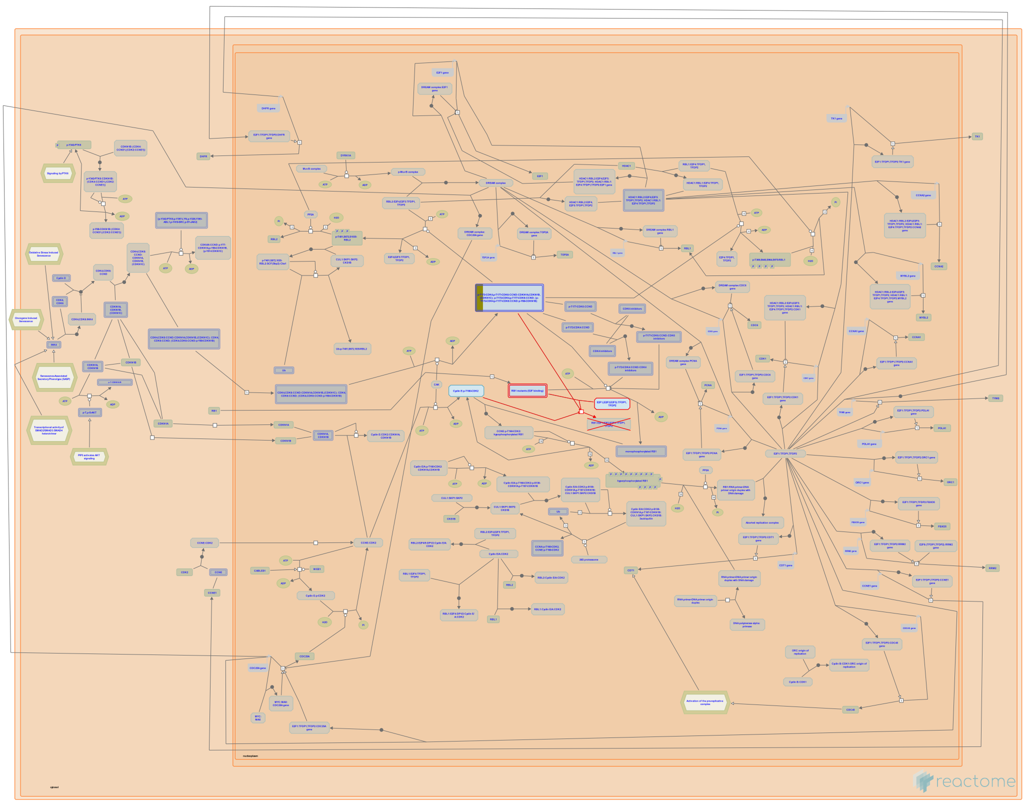
Edit history

Date	Action	Author
2016-06-07	Created	Orlic-Milacic M
2016-06-14	Edited	Orlic-Milacic M
2016-06-14	Authored	Orlic-Milacic M
2016-07-11	Reviewed	Heynen G, Birchmeier W

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GAB1	Q13480

11. Defective binding of RB1 mutants to E2F1,(E2F2, E2F3) (R-HSA-9661069)



Cellular compartments: nucleoplasm.

Diseases: cancer.

This pathway describes impaired binding of RB1 pocket domain mutants to activating E2Fs, E2F1, E2F2 and E2F3 (Templeton et al. 1991, Helin et al. 1993, Otterson et al. 1997, Ji et al. 2004).

References

Kaye FJ, Otterson GA, Chen Wd, Coxon AB & Khleif SN (1997). Incomplete penetrance of familial retinoblastoma linked to germ-line mutations that result in partial loss of RB function. Proc. Natl. Acad. Sci. U.S.A., 94, 12036-40.

Ji P, Rekhtman K, Pagano M, Bloom J, Jiang H, Zhu L & Ichetovkin M (2004). An Rb-Skp2-p27 pathway mediates acute cell cycle inhibition by Rb and is retained in a partial-penetrance Rb mutant . Mol. Cell, 16, 47-58.

Fattaey A, Harlow E & Helin K (1993). Inhibition of E2F-1 transactivation by direct binding of the retinoblastoma protein. Mol. Cell. Biol., 13, 6501-8.

Park SH, Weinberg RA, Lanier L & Templeton DJ (1991). Nonfunctional mutants of the retinoblastoma protein are characterized by defects in phosphorylation, viral oncoprotein association, and nuclear tethering. Proc. Natl. Acad. Sci. U.S.A., 88, 3033-7.

Edit history

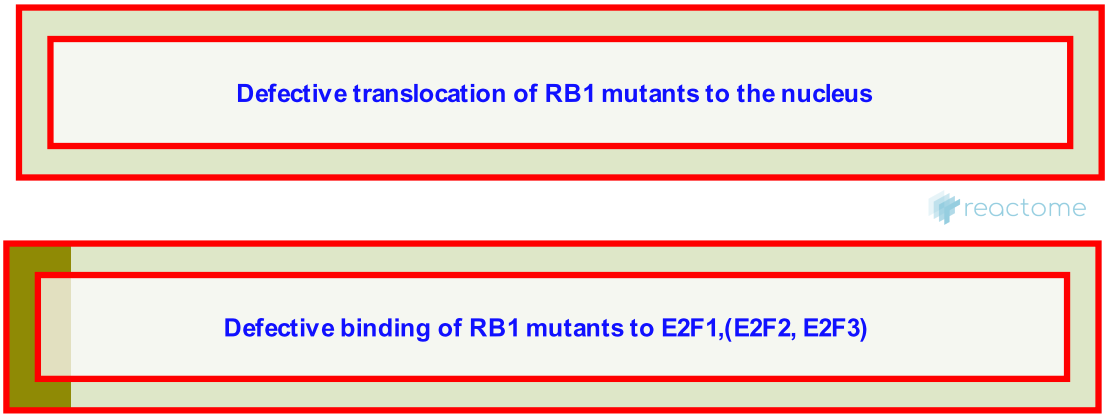
Date	Action	Author
2019-09-13	Created	Orlic-Milacic M
2020-05-07	Authored	Orlic-Milacic M

Date	Action	Author
2020-05-17	Reviewed	Dick FA
2020-05-18	Edited	Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
CCND2	P30279

12. Aberrant regulation of mitotic G1/S transition in cancer due to RB1 defects (R-HSA-9659787)



Diseases: cancer.

RB1 protein, also known as pRB or retinoblastoma protein, is a nuclear protein that plays a major role in the regulation of the G1/S transition during mitotic cell cycle in multicellular eukaryotes. RB1 performs this function by binding to activating E2Fs (E2F1, E2F2 and E2F3), and preventing transcriptional activation of E2F1/2/3 target genes, which include a number of genes involved in DNA synthesis. RB1 binds E2F1/2/3 through the so-called pocket region, which is formed by two parts, pocket domain A (amino acid residues 373-579) and pocket domain B (amino acid residues 640-771). Besides intact pocket domains, RB1 requires an intact nuclear localization signal (NLS) at its C-terminus (amino acid residues 860-876) to be fully functional (reviewed by Classon and Harlow 2002, Dick 2007). Functionally characterized RB1 mutations mostly affect pocket domains A and B and the NLS. RB1 mutations reported in cancer are, however, scattered over the entire RB1 coding sequence and the molecular consequences of the vast majority of these mutations have not been studied (reviewed by Dick 2007).

Many viral oncoproteins inactivate RB1 by competing with E2F1/2/3 for binding to the pocket region of RB1. RB1 protein is targeted by the large T antigen of the Simian virus 40 (SV40), the adenoviral E1A protein, and the E7 protein of oncogenic human papilloma viruses (HPVs) (reviewed by Classon and Harlow 2002).

References

Classon M & Harlow E (2002). The retinoblastoma tumour suppressor in development and cancer. Nat. Rev. Cancer, 2, 910-7.

Dick FA (2007). Structure-function analysis of the retinoblastoma tumor suppressor protein - is the whole a sum of its parts?. Cell Div, 2, 26.

Edit history

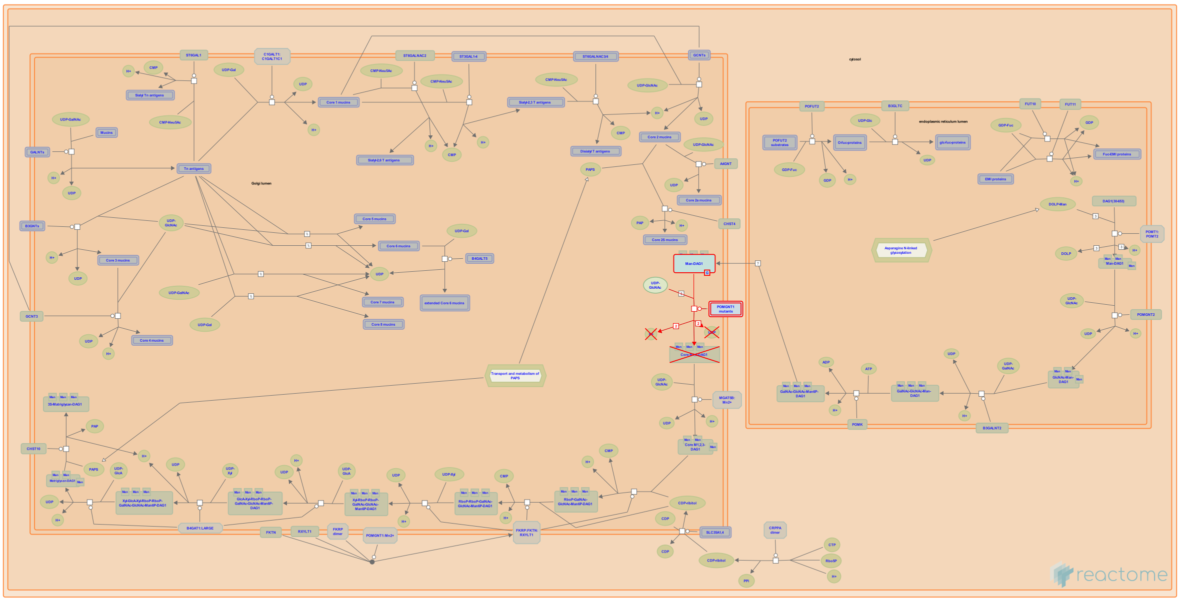
Date	Action	Author
2019-09-05	Created	Orlic-Milacic M
2020-05-07	Authored	Orlic-Milacic M
2020-05-17	Reviewed	Dick FA

Date	Action	Author
2020-05-18	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
CCND2	P30279

13. Defective POMGNT1 causes MDDGA3, MDDGB3 and MDDGC3 (R-HSA-5083628)



Diseases: muscular dystrophy-dystroglycanopathy.

Protein O-linked-mannose beta-1,2-N-acetylglucosaminyltransferase 1 (POMGNT1; CAZy family GT61; MIM:606822) mediates the transfer of N-acetylglucosaminyl (GlcNAc) residues to mannosylated proteins such as mannose-O-serine-dystroglycan (man-O-Ser-DAG1). DAG1 is a cell surface protein that plays an important role in the assembly of the extracellular matrix in muscle, brain, and peripheral nerves by linking the basal lamina to cytoskeletal proteins. Defects in POMGNT1 (MIM:606822) result in disrupted glycosylation of DAG1 and can cause severe congenital muscular dystrophy-dystroglycanopathies ranging from a severe type A3 (MDDGA3; MIM:253280), through a less severe type B3 (MDDGB3; MIM:613151) to a milder type C3 (MDDGC3; MIM:613157) (Bertini et al. 2011, Wells 2013).

References

Wells L (2013). The o-mannosylation pathway: glycosyltransferases and proteins implicated in congenital muscular dystrophy. J. Biol. Chem., 288, 6930-5.

Gualandi F, D'Amico A, Petrini S & Bertini E (2011). Congenital muscular dystrophies: a brief review . Semin Pediatr Neurol, 18, 277-88.

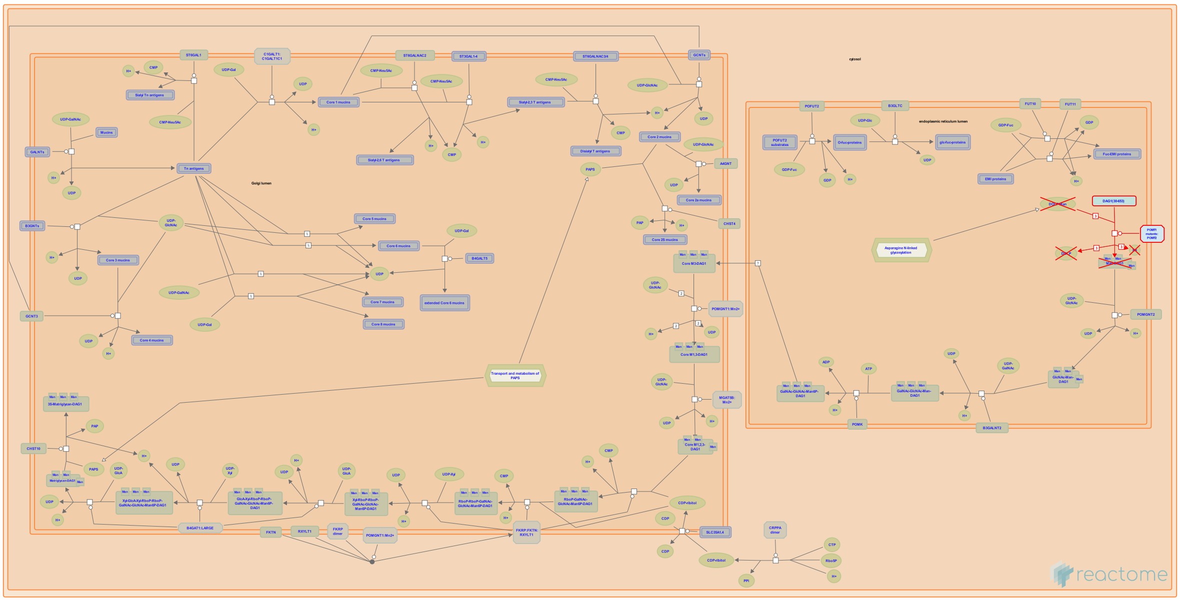
Edit history

Date	Action	Author
2013-11-07	Edited	Jassal B
2013-11-07	Authored	Jassal B
2013-11-07	Created	Jassal B
2015-12-18	Reviewed	Hansen L, Joshi HJ
2023-10-12	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UTRN	P46939	Q14118			

14. Defective POMT1 causes MDDGA1, MDDGB1 and MDDGC1 (R-HSA-5083633)



Diseases: muscular dystrophy-dystroglycanopathy.

Co-expression of both protein O-mannosyl-transferases 1 and 2 (POMT1 and POMT2; CAZy family GT39) is necessary for enzyme activity, that is mediating the transfer of mannosyl residues to the hydroxyl group of serine or threonine residues of proteins such as alpha-dystroglycan (DAG1; MIM:128239). DAG1 is a cell surface protein that plays an important role in the assembly of the extracellular matrix in muscle, brain, and peripheral nerves by linking the basal lamina to cytoskeletal proteins. Defects in POMT1 (MIM:607423) results in defective glycosylation of DAG1 and can cause severe congenital muscular dystrophy-dystroglycanopathies ranging from a severe type A, MDDGA1 (brain and eye abnormalities; MIM:236670), through a less severe type B, MDDGB1 (congenital form with mental retardation; MIM:613155) to a milder type C, MDDGC1 (limb girdle form; MIM:609308) (Bertini et al. 2011, Wells 2013).

References

Wells L (2013). The o-mannosylation pathway: glycosyltransferases and proteins implicated in congenital muscular dystrophy. J. Biol. Chem., 288, 6930-5.

Gualandi F, D'Amico A, Petrini S & Bertini E (2011). Congenital muscular dystrophies: a brief review . Semin Pediatr Neurol, 18, 277-88.

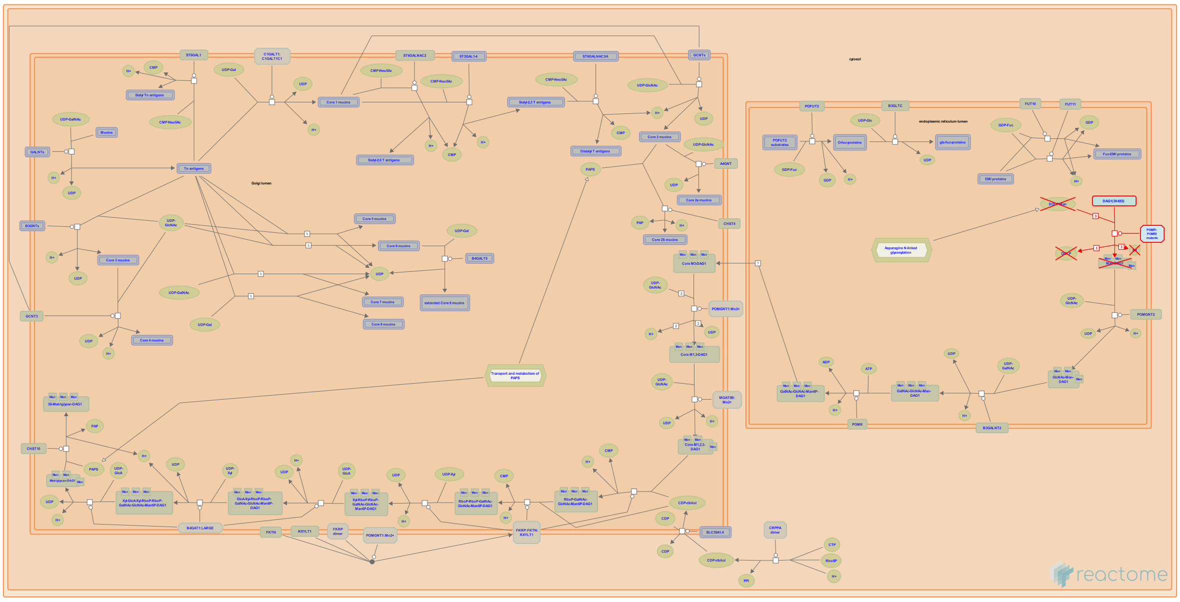
Edit history

Date	Action	Author
2013-11-07	Edited	Jassal B
2013-11-07	Authored	Jassal B
2013-11-07	Created	Jassal B
2015-12-18	Reviewed	Hansen L, Joshi HJ
2023-10-12	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UTRN	P46939	Q14118			

15. Defective POMT2 causes MDDGA2, MDDGB2 and MDDGC2 (R-HSA-5083629)



Diseases: muscular dystrophy-dystroglycanopathy.

Co-expression of both protein O-mannosyl-transferases 1 and 2 (POMT1 and POMT2; CAZy family GT39) is necessary for enzyme activity, that is mediating the transfer of mannosyl residues to the hydroxyl group of serine or threonine residues of proteins such as alpha-dystroglycan (DAG1; MIM:128239). DAG1 is a cell surface protein that plays an important role in the assembly of the extracellular matrix in muscle, brain, and peripheral nerves by linking the basal lamina to cytoskeletal proteins. Defects in POMT2 (MIM:607439) results in defective glycosylation of DAG1 and can cause severe congenital muscular dystrophy dystroglycanopathies ranging from a severe type A, MDDGA2 (brain and eye abnormalities; MIM:613150), through a less severe type B, MDDGB2 (congenital form with mental retardation; MIM:613156) to a milder type C, MDDGC2 (limb girdle form; MIM:603158) (Bertini et al. 2011, Wells 2013).

References

Wells L (2013). The o-mannosylation pathway: glycosyltransferases and proteins implicated in congenital muscular dystrophy. J. Biol. Chem., 288, 6930-5.

Gualandi F, D'Amico A, Petrini S & Bertini E (2011). Congenital muscular dystrophies: a brief review . Semin Pediatr Neurol, 18, 277-88.

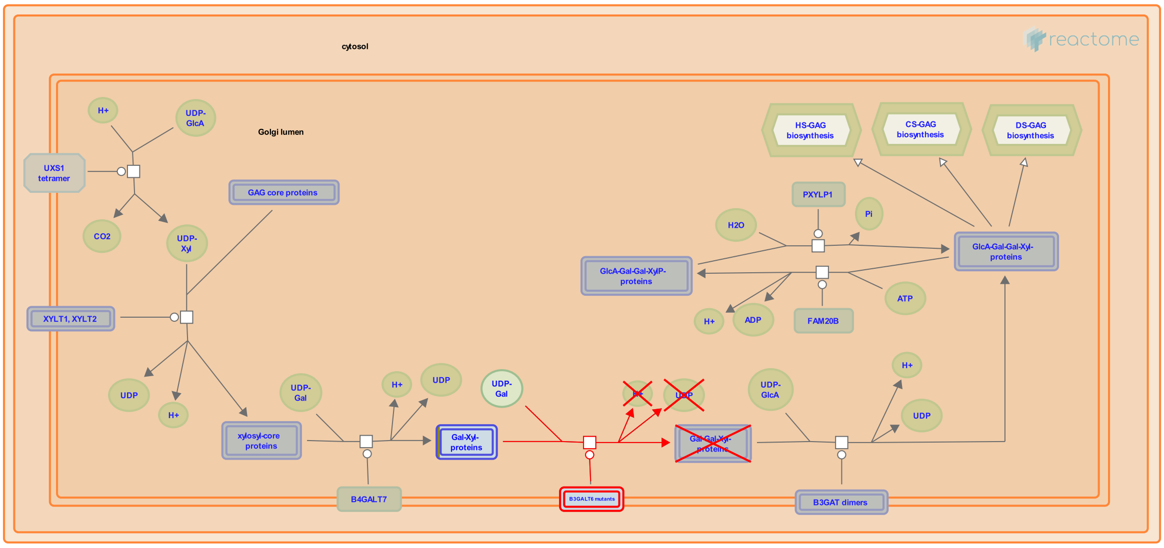
Edit history

Date	Action	Author
2013-11-07	Edited	Jassal B
2013-11-07	Authored	Jassal B
2013-11-07	Created	Jassal B
2015-12-18	Reviewed	Hansen L, Joshi HJ
2023-10-12	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UTRN	P46939	Q14118			

16. Defective B3GALT6 causes EDSP2 and SEMDJL1 (R-HSA-4420332)



Diseases: Ehlers-Danlos syndrome, spondyloepimetaphyseal dysplasia.

The biosynthesis of dermatan sulfate/chondroitin sulfate and heparin/heparan sulfate glycosaminoglycans (GAGs) starts with the formation of a tetrasaccharide linker sequence attached to the core protein. Beta-1,3-galactosyltransferase 6 (B3GALT6) is one of the critical enzymes involved in the formation of this linker sequence. Defects in B3GALT6 causes Ehlers-Danlos syndrome progeroid type 2 (EDSP2; MIM:615349), a severe disorder resulting in a broad spectrum of skeletal, connective tissue and wound healing problems. Defects in B3GALT6 can also cause spondyloepimetaphyseal dysplasia with joint laxity type 1 (SEMDJL1; MIM:271640), characterised by spinal deformaty and lax joints, especially of the hands and respiratory compromise resulting in early death (Nakajima et al. 2013, Malfait et al. 2013).

References

Miyake N, Saitsu H, Iida A, Miyazaki O, Bonafe L, Cavalcanti D, ... Dupuis L (2013). Mutations in B3GALT6, which Encodes a Glycosaminoglycan Linker Region Enzyme, Cause a Spectrum of Skeletal and Connective Tissue Disorders. Am. J. Hum. Genet..

Ebrahimiadib N, Syx D, Hausser I, Symoens S, Malfait F, Gauche C, ... Fournel-Gigleux S (2013). Defective Initiation of Glycosaminoglycan Synthesis due to B3GALT6 Mutations Causes a Pleiotropic Ehlers-Danlos Syndrome-like Connective Tissue Disorder. Am. J. Hum. Genet..

Edit history

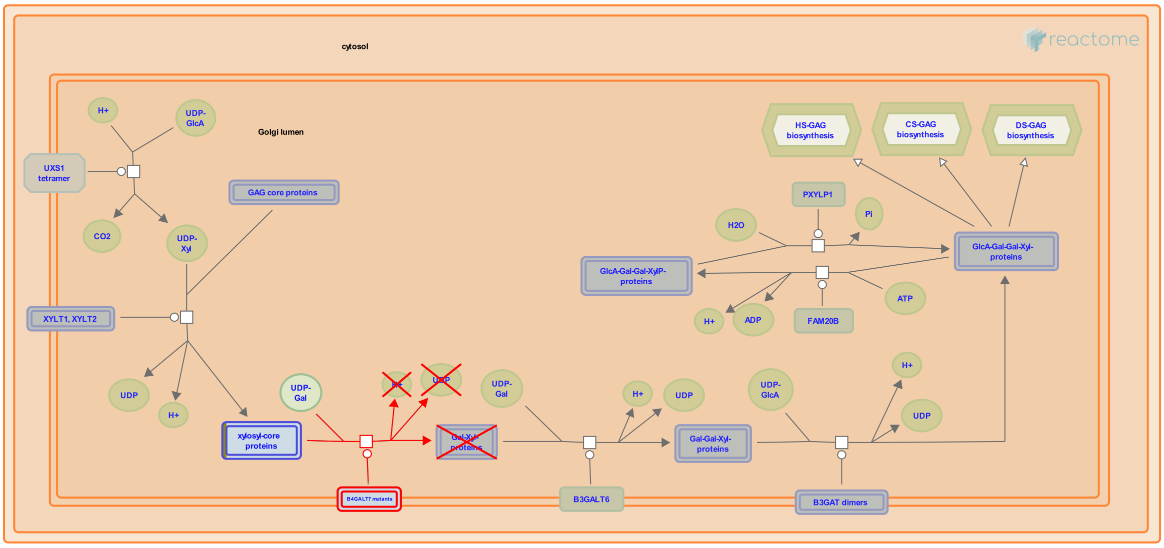
Date	Action	Author
2013-09-03	Edited	Jassal B
2013-09-03	Authored	Jassal B
2013-09-03	Created	Jassal B
2014-07-09	Reviewed	Spillmann D

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GPC6	Q9Y625

Input	UniProt Id

17. Defective B4GALT7 causes EDS, progeroid type (R-HSA-3560783)



Diseases: Ehlers-Danlos syndrome.

Ehlers–Danlos syndrome (EDS) is a group of inherited connective tissue disorders, caused by a defect in the synthesis of collagen types I or III. Abnormal collagen renders connective tissues more elastic. The severity of the mutation can vary from mild to life-threatening. There is no cure and treatment is supportive, including close monitoring of the digestive, excretory and particularly the cardiovascular systems. Defective B4GALT7, a galactosyltransferase important in proteoglycan synthesis, causes the progeroid variant of EDS (MIM:130070). Features include an aged appearance, developmental delay, short stature, generalized osteopenia, defective wound healing, hypermobile joints, hypotonic muscles, and loose but elastic skin (Okajima et al. 1999).

References

Fukumoto S, Urano T, Okajima T & Furukawa K (1999). Molecular basis for the progeroid variant of Ehlers-Danlos syndrome. Identification and characterization of two mutations in galactosyltransferase I gene. J Biol Chem, 274, 28841-4.

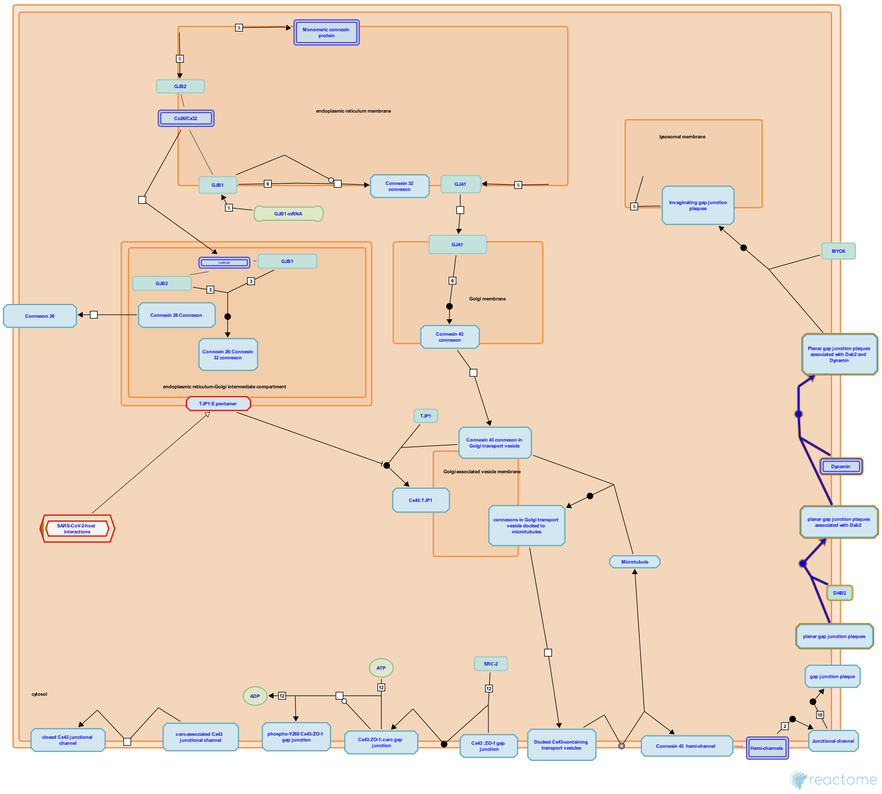
Edit history

Date	Action	Author
2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GPC6	Q9Y625

18. Formation of annular gap junctions (R-HSA-196025)



Gap junction plaque internalization and the disruption cell communication requires a reorganization of Cx molecular interactions. Proteins including Dab-2, AP-2, Dynamin and Myosin VI associate with gap junction plaques permitting the internalisation of plaques after clathrin association (Piehl et al., 2007). Until now, two kinds of annular gap junctions have been described. The first is a small vesicle like structure which permits gap junction plaque renewal without arrest of functionality (Jordan et al., 2001). The second is a large annular structure, composed primarily of the junctional plaques of two adjacent cells (Jordan et al., 2001; Segretain et al., 2004).

References

Gumpert A, Segretain D, Falk MM, Piehl M, Denizot JP & Lehmann C (2007). Internalization of Large Double-Membrane Intercellular Vesicles by a Clathrin-dependent Endocytic Process. Mol Biol Cell.

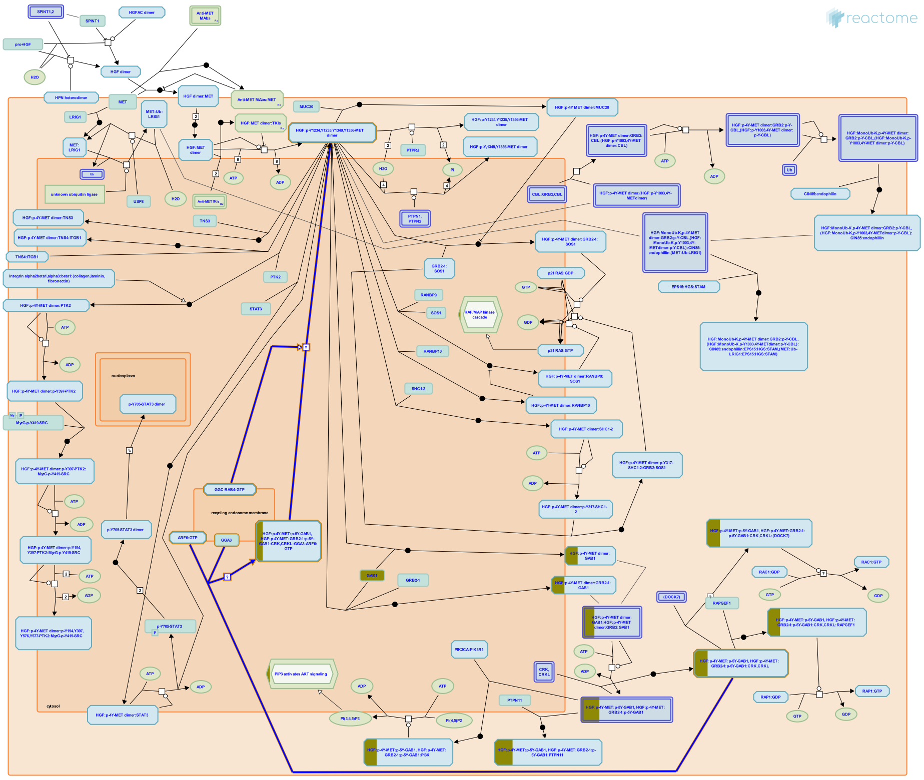
Edit history

Date	Action	Author
2007-01-03	Authored	Gilleron J, Segretain D, Falk MM
2007-04-17	Created	Matthews L
2025-06-09	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SMAD2	Q15796	P98082			

19. MET receptor recycling (R-HSA-8875656)



Activated MET receptor is subject to recycling from the plasma membrane through the endosomal compartment and back to the plasma membrane (Peschard et al. 2001, Hammond et al. 2001, Petrelli et al. 2002). In the recycling process, activated MET receptor is endocytosed, and the GGA3 protein directs it, via a largely unknown mechanism, through the RAB4 positive endosomal compartments back to the plasma membrane (Parachoniak et al. 2011). Endosomal signaling by MET during the recycling process appears to play an important role in sustained activation of ERK1/ERK2 (MAPK3/MAPK1) and STAT3 downstream of MET (Kermorgant and Parker 2008).

References

Kermorgant S & Parker PJ (2008). Receptor trafficking controls weak signal delivery: a strategy used by c-Met for STAT3 nuclear accumulation. J. Cell Biol., 182, 855-63.

Gilestro GF, Migone N, Lanzardo S, Giordano S, Comoglio PM & Petrelli A (2002). The endophilin-CIN85-Cbl complex mediates ligand-dependent downregulation of c-Met. Nature, 416, 187-90.

Urbé S, Clague MJ, Hammond DE & Vande Woude GF (2001). Down-regulation of MET, the receptor for hepatocyte growth factor. Oncogene, 20, 2761-70.

Abella JV, Park M, Luo Y, Parachoniak CA & Keen JH (2011). GGA3 functions as a switch to promote Met receptor recycling, essential for sustained ERK and cell migration. Dev. Cell, 20, 751-63.

Band H, Peschard P, Naujokas MA, Fournier TM, Park M, Lamorte L & Langdon WY (2001). Mutation of the c-Cbl TKB domain binding site on the Met receptor tyrosine kinase converts it into a transforming protein. Mol. Cell, 8, 995-1004.

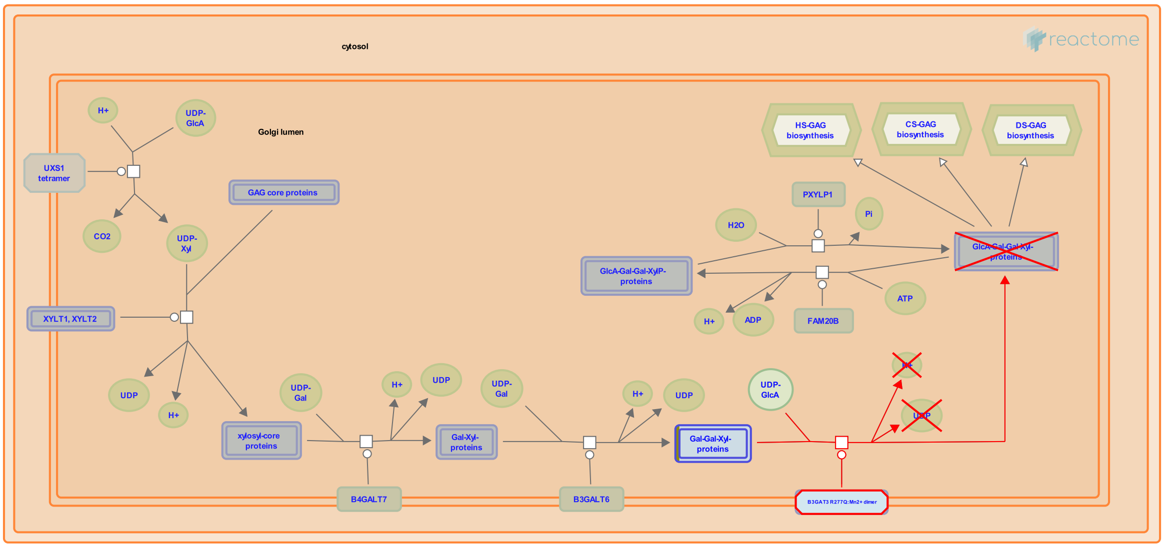
Edit history

Date	Action	Author
2016-06-08	Created	Orlic-Milacic M
2016-06-14	Edited	Orlic-Milacic M
2016-06-14	Authored	Orlic-Milacic M
2016-07-11	Reviewed	Heynen G, Birchmeier W

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GAB1	Q13480

20. Defective B3GAT3 causes JDSSDHD (R-HSA-3560801)



Diseases: Larsen syndrome, congenital heart defect.

Galactosylgalactosylxylosylprotein 3-beta-glucuronosyltransferases1, 2 and 3 (B3GAT1-3) are involved in forming the linker tetrasaccharide present in heparan sulfate and chondroitin sulfate. Defects in B3GAT3 cause multiple joint dislocations, short stature, craniofacial dysmorphism, and congenital heart defects (JDSSDHD; MIM:245600). This is an autosomal recessive disease characterized by dysmorphic facies, elbow, hip and knee dislocations, clubfeet, short stature and cardiovascular defects (Steel & Kohl 1972, Bonaventure et al. 1992, Baasanjav et al. 2011). JDSSDHD has phenotypic similarities to Larsen syndrome (Larsen et al. 1950).

References

Becker C, Fischer B, Horn D, Hashiguchi T, Stricker S, Aziz SA, ... Baasanjav S (2011). Faulty initiation of proteoglycan synthesis causes cardiac and joint defects. Am. J. Hum. Genet., 89, 15-27.

LARSEN LJ, SCHOTTSTAEDT ER & BOST FC (1950). Multiple congenital dislocations associated with characteristic facial abnormality. J. Pediatr., 37, 574-81.

Steel HH & Kohl EJ (1972). Multiple congenital dislocations associated with other skeletal anomalies (Larsen's syndrome) in three siblings. J Bone Joint Surg Am, 54, 75-82.

Lasselin C, Mellier J, Bonaventure J, MAROTEAUX P & Cohen-Solal L (1992). Linkage studies of four fibrillar collagen genes in three pedigrees with Larsen-like syndrome. J. Med. Genet., 29, 465-70.

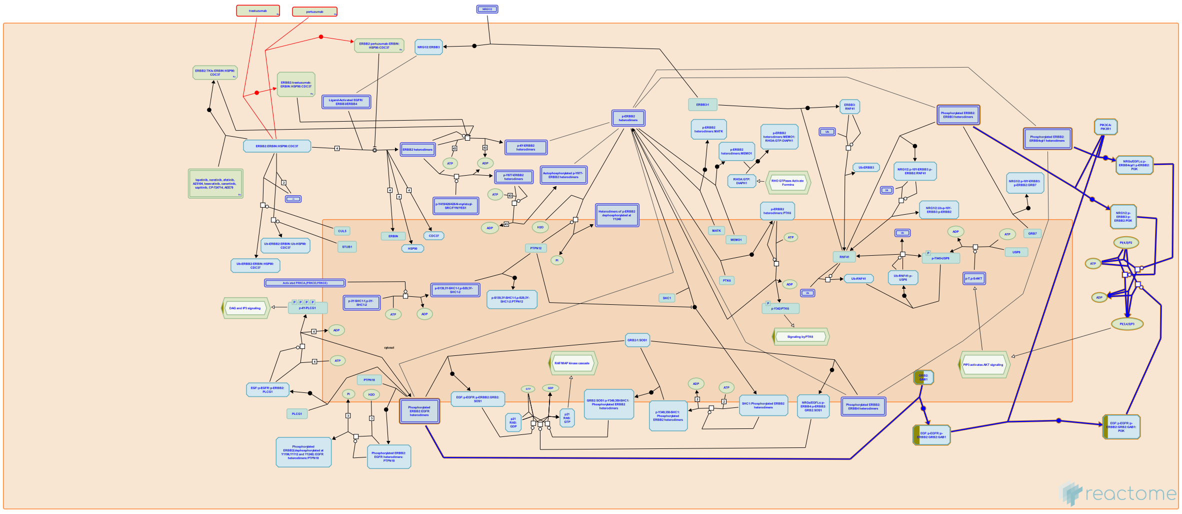
Edit history

Date	Action	Author
2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GPC6	Q9Y625

21. PI3K events in ERBB2 signaling (R-HSA-1963642)



ERBB2:ERBB3 and ERBB2:ERBB4cyt1 heterodimers activate PI3K signaling by direct binding of PI3K regulatory subunit p85 (Yang et al. 2007, Cohen et al. 1996, Kaushansky et al. 2008) to phosphorylated tyrosine residues in the C-tail of ERBB3 (Y1054, Y1197, Y1222, Y1224, Y1276 and Y1289) and ERBB4 CYT1 isoforms (Y1056 in JM-A CYT1 isoform and Y1046 in JM-B CYT1 isoform). Regulatory subunit p85 subsequently recruits catalytic subunit p110 of PI3K, resulting in the formation of active PI3K, conversion of PIP2 to PIP3, and PIP3-mediated activation of AKT signaling (Junttila et al. 2009, Kainulainen et al. 2000). Heterodimers of ERBB2 and EGFR recruit PI3K indirectly, through GRB2:GAB1 complex (Jackson et al. 2004), which again leads to PIP3-mediated activation of AKT signaling.

References

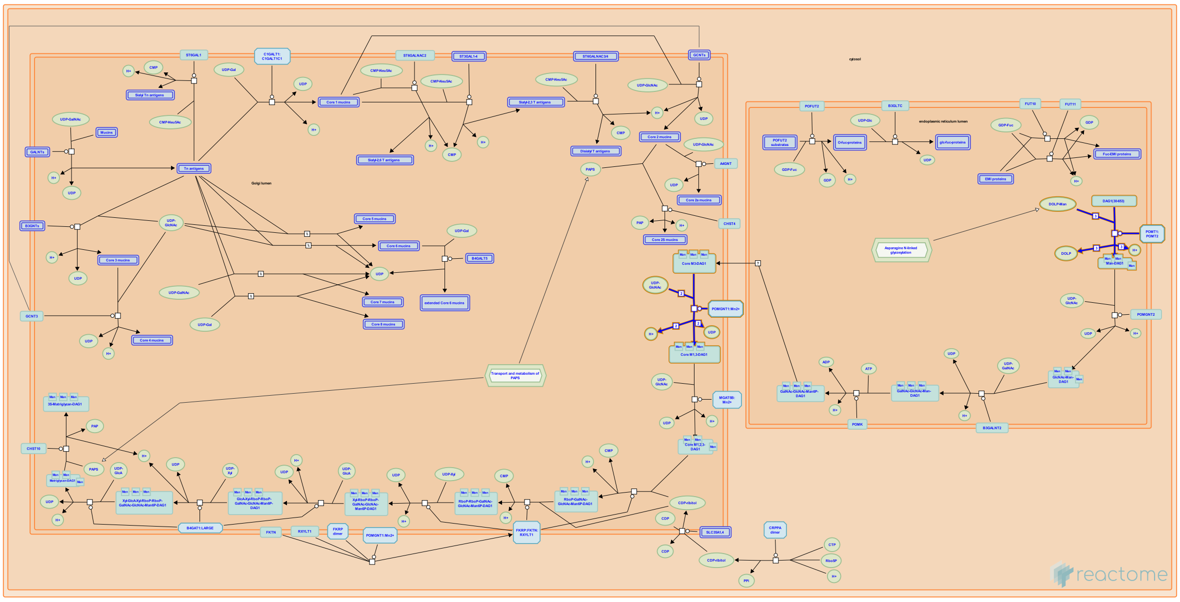
Edit history

Date	Action	Author
2011-11-04	Authored	Orlic-Milacic M
2011-11-04	Created	Orlic-Milacic M
2011-11-07	Edited	Matthews L, D'Eustachio P
2011-11-11	Reviewed	Xu W, Neckers LM

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GAB1	Q13480

22. DAG1 core M1 glycosylations (R-HSA-8932506)



Protein-linked O-mannose glycans are initiated by the covalent attachment of mannose to serine and threonine residues in the protein via alpha-linkages. Core M1 glycans consist of all O-mannose glycans in which O-mannose is extended with beta-1,2-linked N-acetylglucosamine (GlcNAc) but not with beta-1,6-linked GlcNAc. M1 structures do not directly bind key extracellular components but are essential as precursors for M2 structures, and they support the maturation of M3 glycans (Praissman & Wells 2014; Endo, 2019).

References

Wells L & Praissman JL (2014). Mammalian O-mannosylation pathway: glycan structures, enzymes, and protein substrates. Biochemistry, 53, 3066-78.

Endo T (2019). Mammalian O-mannosyl glycans: Biochemistry and glycopathology. Proc Jpn Acad Ser B Phys Biol Sci, 95, 39-51.

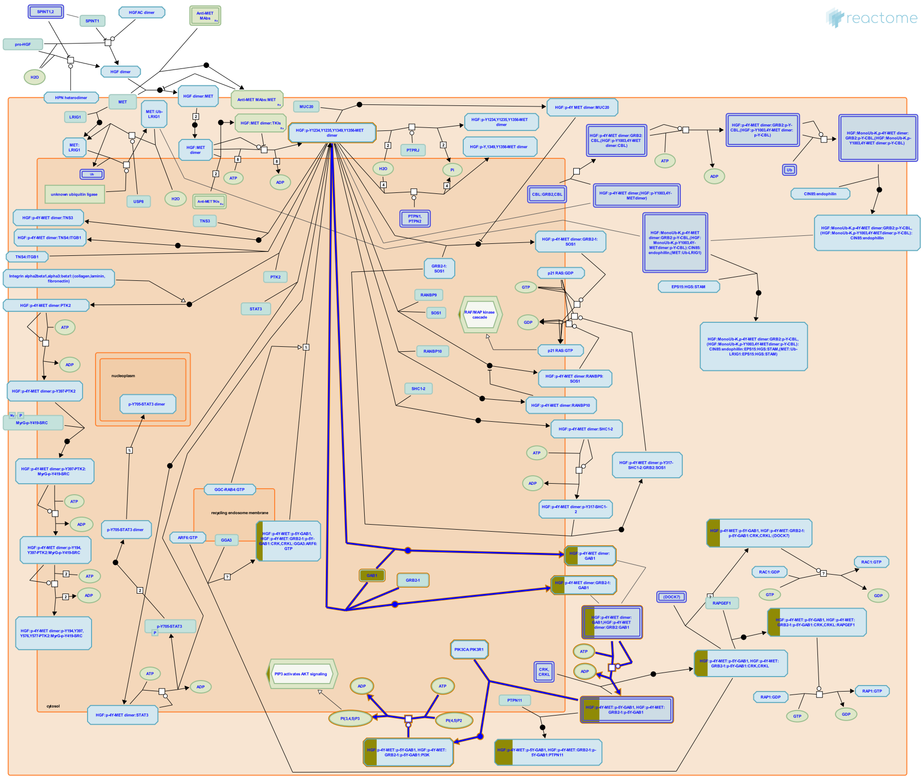
Edit history

Date	Action	Author
2016-07-26	Edited	Jassal B
2016-07-26	Authored	Jassal B
2016-07-26	Created	Jassal B
2025-05-01	Reviewed	Zaru R
2025-06-09	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UTRN	P46939	Q14118			

23. MET activates PI3K/AKT signaling (R-HSA-8851907)



MET binds and phosphorylates the adapter protein GAB1, thus creating a docking site for the regulatory subunit PIK3R1 of the PI3K complex. Recruitment of PI3K to MET-bound phosphorylated GAB1 results in PI3K activation, production of PIP3, and stimulation of downstream AKT signaling (Rodrigues et al. 2000, Schaeper et al. 2000).

References

Kempkes B, Fuchs KP, Birchmeier W, Sachs M, Gehring NH & Schaeper U (2000). Coupling of Gab1 to c-Met, Grb2, and Shp2 mediates biological responses. J. Cell Biol., 149, 1419-32.

Falasca M, Rodrigues GA, Schlessinger J, Ong SH & Zhang Z (2000). A novel positive feedback loop mediated by the docking protein Gab1 and phosphatidylinositol 3-kinase in epidermal growth factor receptor signaling. Mol Cell Biol, 20, 1448-59.

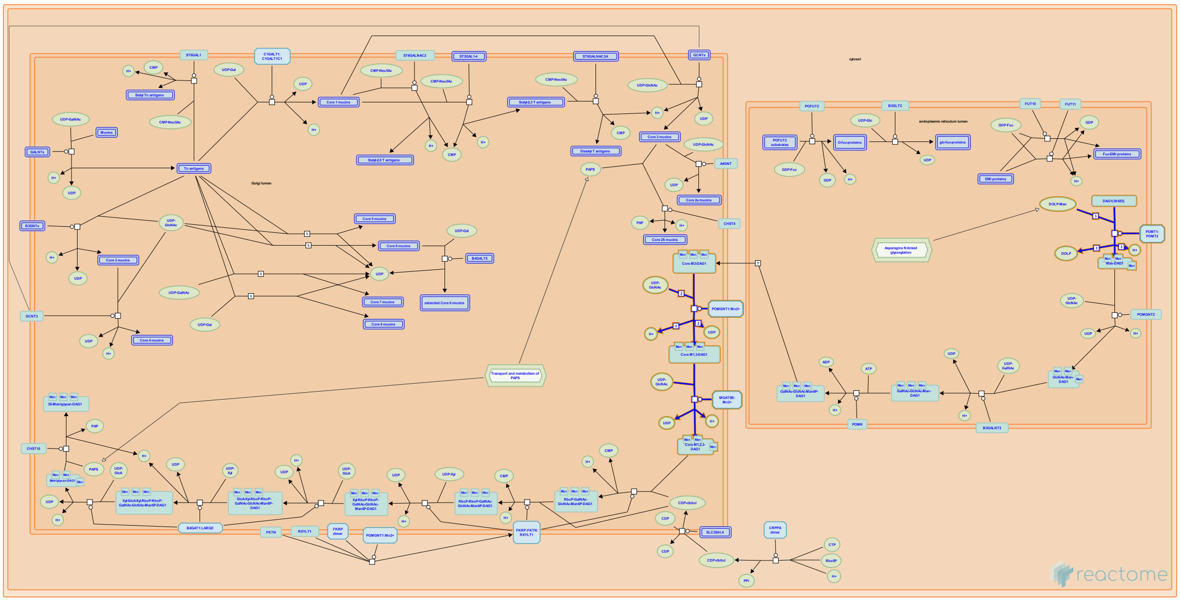
Edit history

Date	Action	Author
2016-01-12	Created	Orlic-Milacic M
2016-06-14	Edited	Orlic-Milacic M
2016-06-14	Authored	Orlic-Milacic M
2016-07-11	Reviewed	Heynen G, Birchmeier W

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GAB1	Q13480

24. DAG1 core M2 glycosylations (R-HSA-8932504)



Core M2 glycans are initiated by beta-1,6-linked GlcNAc extension of core M1 glycans. These structures are primarily found in brain and prostate tissue and account for no more than 5% of brain protein-linked O-glycans. M2 structures play a role in neural cell adhesion and migration (Praissman & Wells 2014; Endo, 2019).

References

Wells L & Praissman JL (2014). Mammalian O-mannosylation pathway: glycan structures, enzymes, and protein substrates. Biochemistry, 53, 3066-78.

Endo T (2019). Mammalian O-mannosyl glycans: Biochemistry and glycopathology. Proc Jpn Acad Ser B Phys Biol Sci, 95, 39-51.

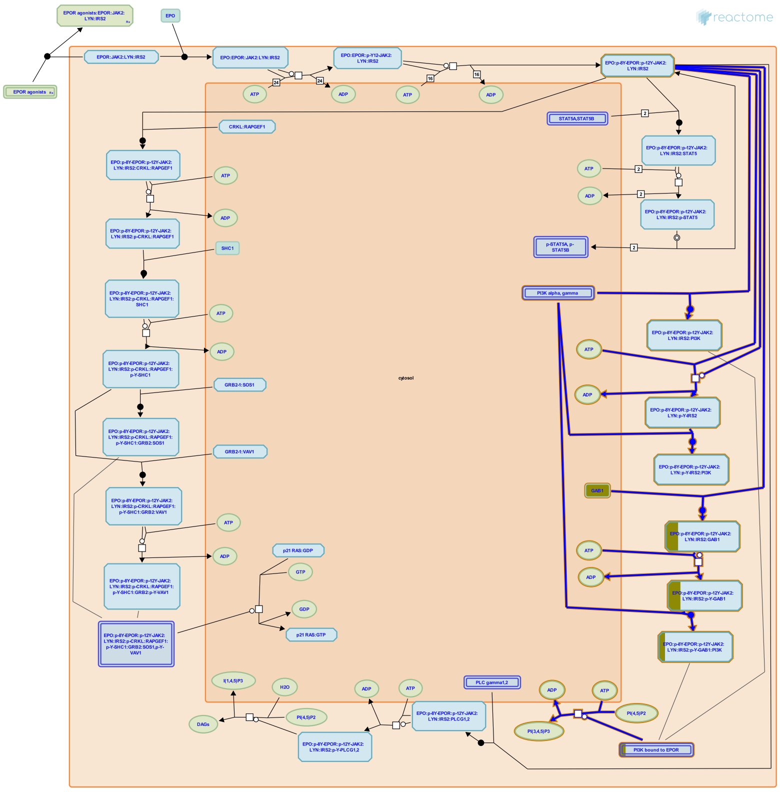
Edit history

Date	Action	Author
2016-07-26	Edited	Jassal B
2016-07-26	Authored	Jassal B
2016-07-26	Created	Jassal B
2025-05-01	Reviewed	Zaru R
2025-06-09	Modified	Weiser JD

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UTRN	P46939	Q14118			

25. Erythropoietin activates Phosphoinositide-3-kinase (PI3K) (R-HSA-9027276)



PI3K can bind the activated EPO receptor (EPOR) by three different mechanisms: direct binding to phospho-Y479 of the EPOR, indirect binding via phosphorylated IRS2 bound to the EPOR, and indirect binding via phosphorylated GAB1 bound to the EPOR (Bouscary et al. 2003, Schmidt et al. 2004, reviewed in Kuhrt and Wojchowski 2015). PI3K phosphorylates phosphatidylinositol 4,5-bisphosphate to yield phosphatidylinositol 3,4,5-trisphosphate which recruits AKT1 to the membrane.

References

Wojchowski DM & Kuhrt D (2015). Emerging EPO and EPO receptor regulators and signal transducers. Blood, 125, 3536-41.

Claessens YE, Mayeux P, Fontenay-Roupie M, Bouscary D, Pene F, Lacombe C, ... Gisselbrecht S (2003). Critical role for PI 3-kinase in the control of erythropoietin-induced erythroid progenitor proliferation. Blood, 101, 3436-43.

Schmidt EK, Feller SM & Fichelson S (2004). PI3 kinase is important for Ras, MEK and Erk activation of Epo-stimulated human erythroid progenitors. BMC Biol., 2, 7.

Edit history

Date	Action	Author
2017-10-29	Edited	May B
2017-10-29	Authored	May B
2017-10-29	Created	May B
2018-08-14	Reviewed	McGraw KL

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GAB1	Q13480

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

10 of the submitted entities were found, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCND2	P30279	GAB1	Q13480	GPC6	Q9Y625
MBOAT1	Q6ZNC8	PPP1R12B	O60237	SMAD2	Q15796
SYT10	Q6XYQ8	TNFSF4	P23510	UTRN	P46939
ZNF875	P10072				

Input	Ensembl Id
SYT10	ENSG00000110975

Interactors (11)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCND2	P30279	P11802	GAB1	Q13480	Q06124
GPC6	Q9Y625	P32249	MAK	P20794	Q9UM11
PPP1R12B	O60237-2	Q14957	RBPMS	Q93062	O94955
SMAD2	Q15796	P98082	SMCO4	Q9NRQ5	P25025
TNFSF4	P23510	P43489	UTRN	P46939	Q14118
ZNF488	Q96MN9-2	Q00403			

7. Identifiers not found

These 8 identifiers were not found neither mapped to any entity in Reactome.

DNAH9	EXPH5	KCNJ13	KCNQ1OT1	LRRC2	PART1	PHACTR1	ZBED3